

Cybersecurity & Ethical Hacking Internship Program

Task 1 – Foundations of Cybersecurity

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Course: B.Tech Cybersecurity

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Task 1 Practical Lab Report

This report documents the cybersecurity fundamentals, Linux and networking concepts, cryptography exercises, and controlled virtual lab work completed for Task 1.

1. Objective

Build foundational knowledge of cybersecurity and gain practical familiarity with Linux, networking, cryptography, and basic security tools in a controlled virtual laboratory.

2. Cybersecurity Basics

CIA Triad: Confidentiality, Integrity, and Availability. **Threats:** phishing, malware, DDoS, SQL injection, brute force, and ransomware. **Attack vectors:** social engineering, wireless attacks, and insider threats.

3. Lab Environment

Oracle VirtualBox was used with Kali Linux as the security testing machine and Metasploitable2 as an intentionally vulnerable target. A private Host-Only network was used for controlled communication.

4. Linux Fundamentals

Practiced `pwd`, `ls`, `cd`, `mkdir`, `touch`, `rm`, `rmdir`, `chmod`, `chown`, `apt`, `dpkg`, `ifconfig`, `ping`, `netstat`, and `traceroute` for navigation, file management, permissions, package management, and network troubleshooting.

5. Networking Basics

Studied the OSI and TCP/IP models, DNS, HTTP/HTTPS, IP addressing, subnetting, and NAT. Connectivity between the lab machines was tested using ICMP ping.

6. Cryptography Basics

Studied symmetric and asymmetric encryption, MD5 and SHA-256 hashing, digital certificates, and SSL/TLS. OpenSSL was used to encrypt and decrypt a text file with AES-256-CBC and verify the recovered plaintext.

7. Security Tool Familiarization

Wireshark was used to capture ICMP traffic. Nmap was used for basic network scanning. Burp Suite was used to observe HTTP requests. Netcat was reviewed as a network debugging and communication utility.

8. Practical Results

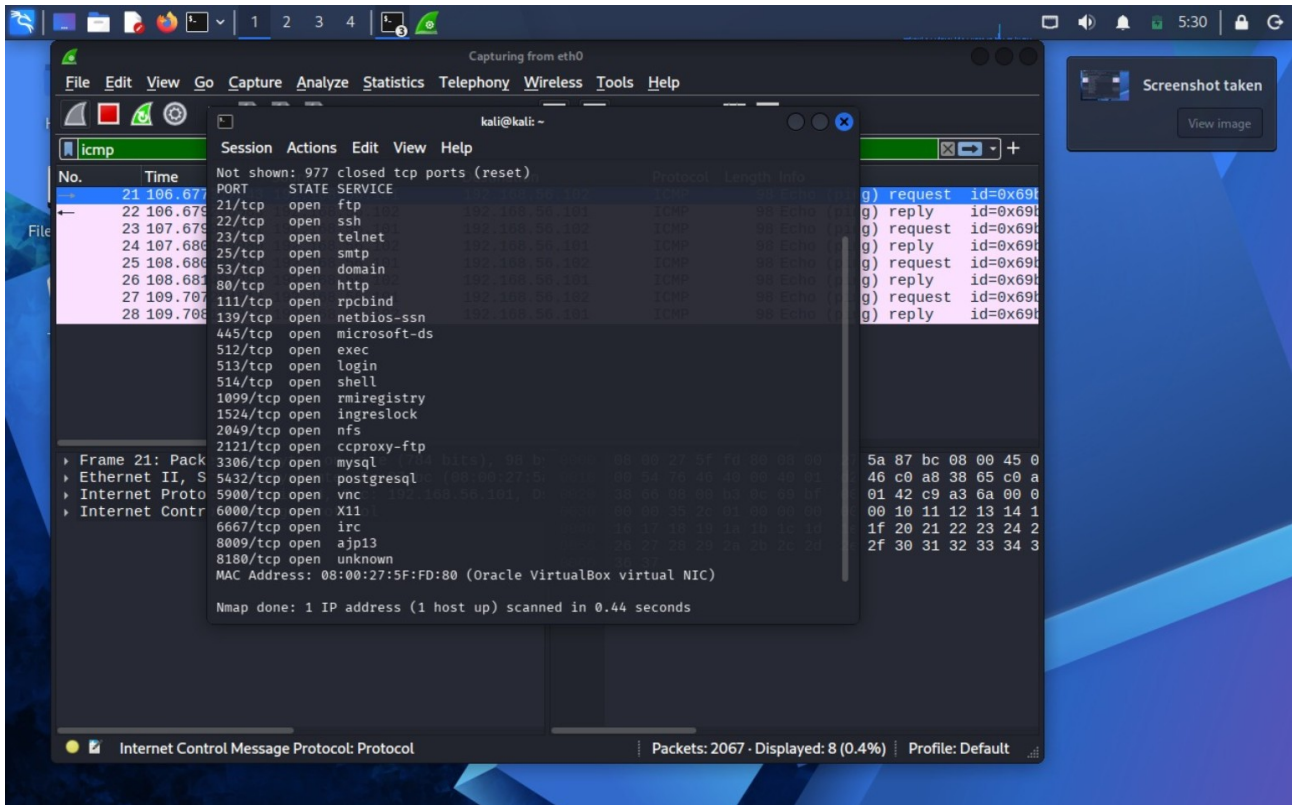
Successfully demonstrated lab connectivity, ICMP packet capture, Linux command practice, basic network scanning, HTTP traffic inspection, hashing, and OpenSSL encryption/decryption.

9. Conclusion

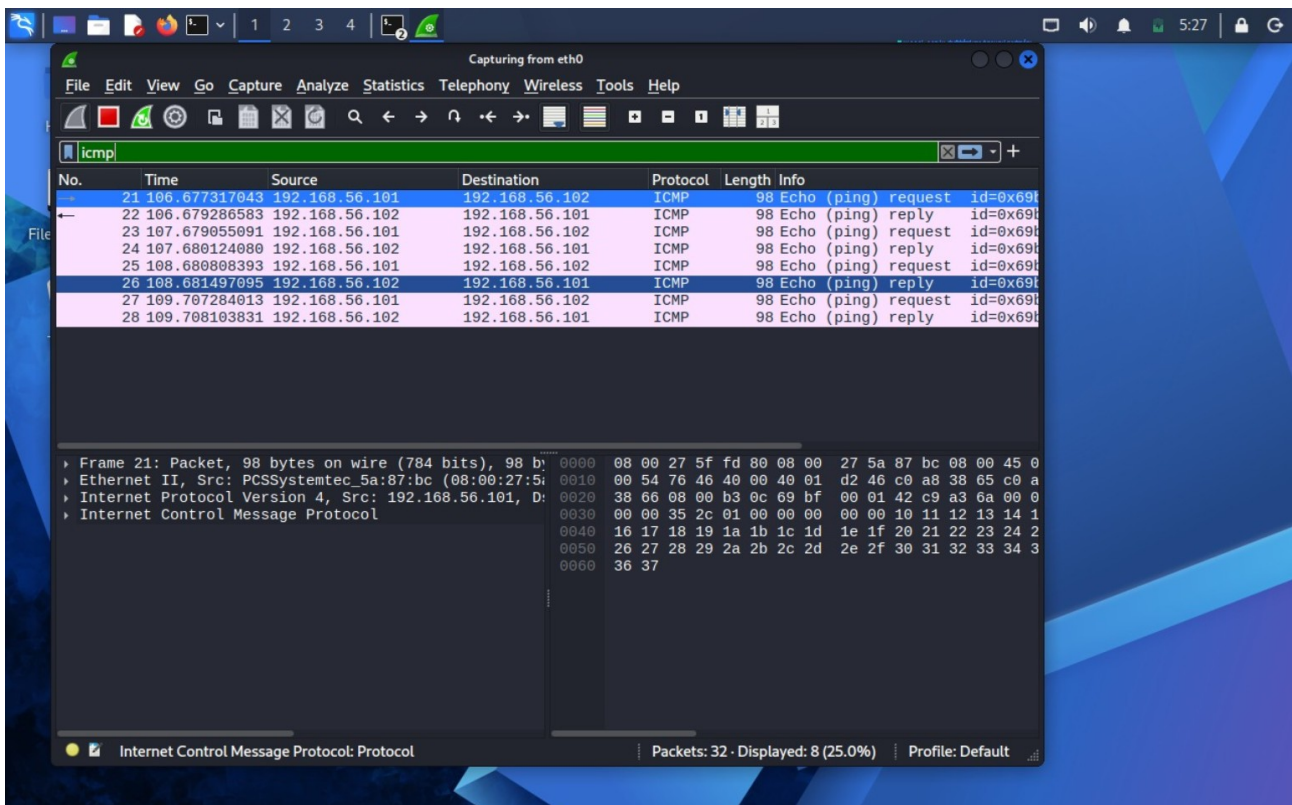
Task 1 connected cybersecurity theory with hands-on practice in a controlled lab and provided foundational experience in Linux, networking, packet analysis, cryptography, and security tools.

10. Lab Screenshots

Kali Linux Lab Evidence



Wireshark ICMP Capture



Metasploitable2 ifconfig

